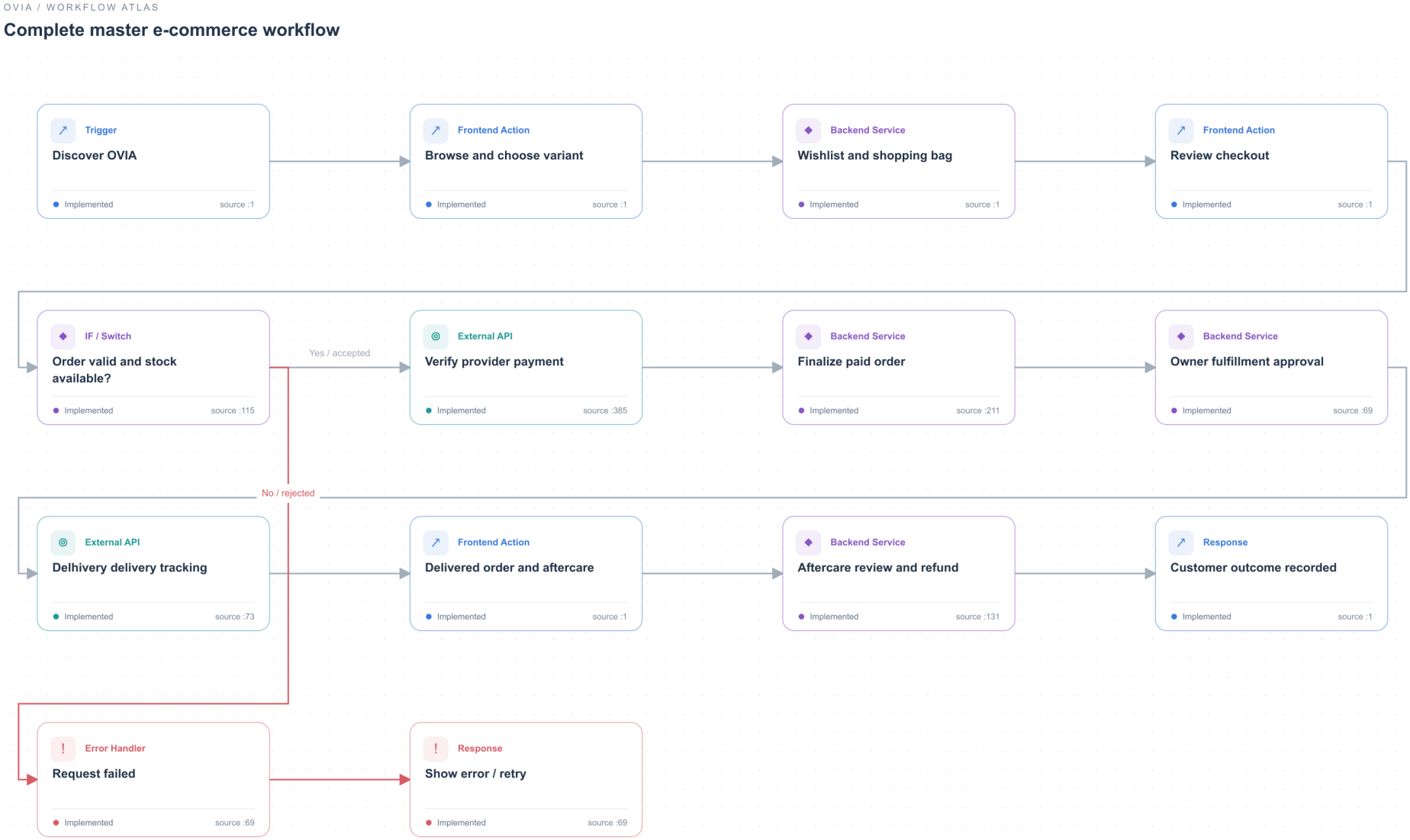
Complete master e-commerce workflow

Architecture overview. Readable node and connection sheets follow.



Complete master e-commerce workflow

Readable detail sheet 1/4. Node numbers and page references preserve every directed connection.

#01 / Trigger / Implemented

Discover OVIA

Home, navigation and bag / luggage storefront.

src/app/page.tsx:1

Next -> #2 (p2)

#02 / Frontend Action / Implemented

Browse and choose variant

Product listing, detail, search and filter flows.

src/components/product-detail.tsx:1

Next -> #3 (p2)

#03 / Backend Service / Implemented

Wishlist and shopping bag

Frontend provider sends cart and wishlist mutations.

src/components/store-provider.tsx:1

Next -> #4 (p2)

#04 / Frontend Action / Implemented

Review checkout

Address, PIN serviceability, shipping method and quote.

src/app/checkout/page.tsx:1

Next -> #5 (p3)

Complete master e-commerce workflow

Readable detail sheet 2/4. Node numbers and page references preserve every directed connection.

#05 / IF / Switch / Implemented

Order valid and stock available?

Idempotent order and stock reservation transaction.

src/server/commerce.ts:115

Yes / accepted -> #6 (p3); No / rejected -> #13 (p5)

#06 / External API / Implemented

Verify provider payment

Razorpay secure checkout; PayPal sandbox is an alternate flow. Runtime configuration and connectivity are Unverified.

src/server/razorpay-checkout.ts:385

Next -> #7 (p3)

#07 / Backend Service / Implemented

Finalize paid order

Confirmed paid state, replay event and mail queue.

src/server/commerce.ts:211

Next -> #8 (p3)

#08 / Backend Service / Implemented

Owner fulfillment approval

Verified payment precedes courier shipment and pickup.

src/server/logistics.ts:69

Next -> #9 (p4)

Complete master e-commerce workflow

Readable detail sheet 3/4. Node numbers and page references preserve every directed connection.

#09 / External API / Implemented

Delhivery delivery tracking

Courier API / authenticated webhook updates stored shipping. Runtime configuration and connectivity are Unverified.

src/lib/shipping/service.ts:73

Next -> #10 (p4)

#10 / Frontend Action / Implemented

Delivered order and aftercare

Customer tracks delivery, submits reviews or opens cancellation / return / refund requests.

src/components/account-order-details.tsx:1

Next -> #11 (p4)

#11 / Backend Service / Implemented

Aftercare review and refund

Tickets are reviewed; gateway refunds follow the separate authorized refund flow.

src/server/care-order-requests.ts:131

Next -> #12 (p4)

#12 / Response / Implemented

Customer outcome recorded

Display delivery, review request, cancellation and provider refund evidence.

src/components/account-orders-returns.tsx:1

END: no outgoing connection

Complete master e-commerce workflow

Readable detail sheet 4/4. Node numbers and page references preserve every directed connection.

#13 / Error Handler / Implemented

Request failed

Validation, access, conflict or service errors are returned to the caller. Status and error shape depend on the specific route.

src/server/http.ts:69

Next -> #14 (p5)

#14 / Response / Implemented

Show error / retry

Failure end state. Caller receives the actual error; a retry may be offered by the UI.

src/server/http.ts:69

END: no outgoing connection

Complete master e-commerce workflow

Complete directed connection register / no omitted success, failure or proposed edges

#01 Discover OVIA -> #02 Browse and choose variant | Next

#02 Browse and choose variant -> #03 Wishlist and shopping bag | Next

#03 Wishlist and shopping bag -> #04 Review checkout | Next

#04 Review checkout -> #05 Order valid and stock available? | Next

#05 Order valid and stock available? -> #06 Verify provider payment | Yes / accepted

#06 Verify provider payment -> #07 Finalize paid order | Next

#07 Finalize paid order -> #08 Owner fulfillment approval | Next

#08 Owner fulfillment approval -> #09 Delhivery delivery tracking | Next

#09 Delhivery delivery tracking -> #10 Delivered order and aftercare | Next

#10 Delivered order and aftercare -> #11 Aftercare review and refund | Next

#11 Aftercare review and refund -> #12 Customer outcome recorded | Next

#05 Order valid and stock available? -> #13 Request failed | No / rejected | FAILURE

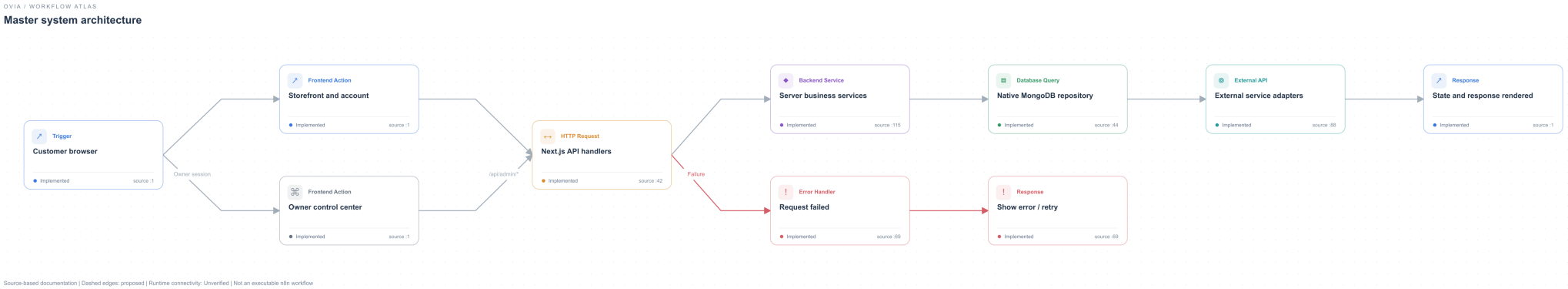
#13 Request failed -> #14 Show error / retry | Next | FAILURE

Note: Source implementation is documented; deployed configuration and provider connectivity are not verified.

Note: This master is a process composition of the individual evidence-mapped workflows, not one executable handler.

Master system architecture

Architecture overview. Readable node and connection sheets follow.



Master system architecture

Readable detail sheet 1/3. Node numbers and page references preserve every directed connection.

#01 / Trigger / Implemented

Customer browser

Next.js storefront shell with customer-facing pages and client components.

src/app/layout.tsx:1

Next -> #2 (p2); Owner session -> #10 (p4)

#02 / Frontend Action / Implemented

Storefront and account

Catalog, bag, checkout and account state coordinate client requests.

src/components/store-provider.tsx:1

Next -> #3 (p2)

#03 / HTTP Request / Implemented

Next.js API handlers

86 route files, including explicit dispatch branches and provider webhooks.

src/app/api/v1/[[...path]]/route.ts:42

Next -> #4 (p2); Failure -> #8 (p3)

#04 / Backend Service / Implemented

Server business services

Commerce, accounts, logistics, mail, support, marketing and reports.

src/server/commerce.ts:115

Next -> #5 (p3)

Master system architecture

Readable detail sheet 2/3. Node numbers and page references preserve every directed connection.

#05 / Database Query / Implemented

Native MongoDB repository

MongoClient pool; repository access and withTransaction for coordinated writes.

src/server/db.ts:44

Next -> #6 (p3)

#06 / External API / Implemented

External service adapters

Razorpay, PayPal sandbox, Delhivery, Resend, Algolia, OAuth, Gmail, translation and DeepSeek. Runtime configuration and connectivity are Unverified.

src/server/integrations-service.ts:88

Next -> #7 (p3)

#07 / Response / Implemented

State and response rendered

Safe route responses update customer and owner views.

src/components/store-provider.tsx:1

END: no outgoing connection

#08 / Error Handler / Implemented

Request failed

Validation, access, conflict or service errors are returned to the caller. Status and error shape depend on the specific route.

src/server/http.ts:69

Next -> #9 (p4)

Master system architecture

Readable detail sheet 3/3. Node numbers and page references preserve every directed connection.

#09 / Response / Implemented

Show error / retry

Failure end state. Caller receives the actual error; a retry may be offered by the UI.

src/server/http.ts:69

END: no outgoing connection

#10 / Frontend Action / Implemented

Owner control center

Server-protected owner operations and API-specific authorization.

src/components/admin/admin-shell.tsx:1

/api/admin/* -> #3 (p2)

Master system architecture

Complete directed connection register / no omitted success, failure or proposed edges

#01 Customer browser -> #02 Storefront and account | Next

#02 Storefront and account -> #03 Next.js API handlers | Next

#03 Next.js API handlers -> #04 Server business services | Next

#04 Server business services -> #05 Native MongoDB repository | Next

#05 Native MongoDB repository -> #06 External service adapters | Next

#06 External service adapters -> #07 State and response rendered | Next

#03 Next.js API handlers -> #08 Request failed | Failure | FAILURE

#08 Request failed -> #09 Show error / retry | Next | FAILURE

#01 Customer browser -> #10 Owner control center | Owner session

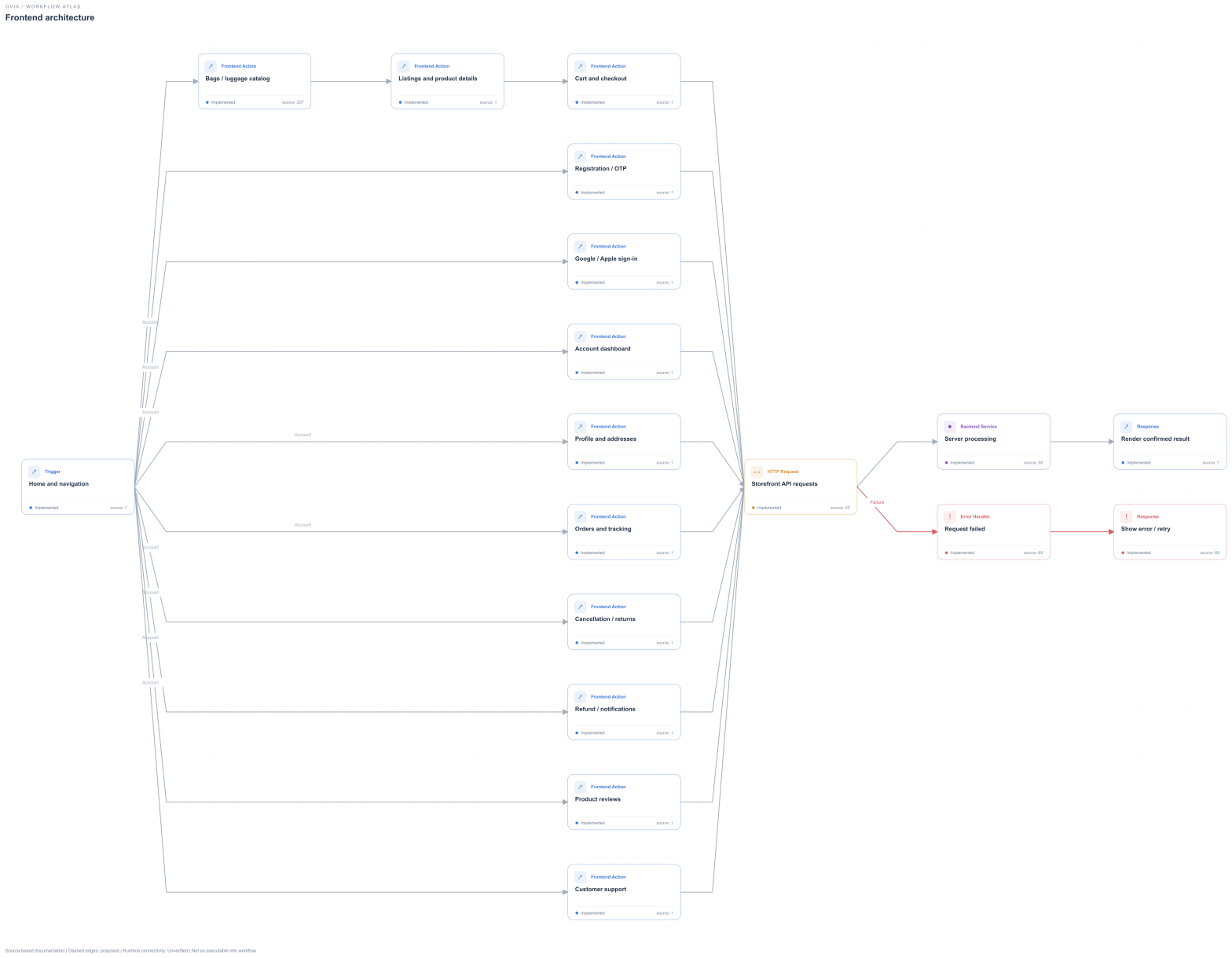
#10 Owner control center -> #03 Next.js API handlers | /api/admin/*

Note: Source implementation is documented; deployed configuration and provider connectivity are not verified.

Note: MongoDB Atlas deployment topology is Unverified. The code proves a MongoDB connection and database selection, not specific cluster tiers or regions.

Frontend architecture

Architecture overview. Readable node and connection sheets follow.



Frontend architecture

Readable detail sheet 1/5. Node numbers and page references preserve every directed connection.

#01 / Trigger / Implemented

Home and navigation

Homepage and shared header / navigation.

src/app/page.tsx:1

Next -> #2 (p2); Account -> #10 (p4); Account -> #11 (p4); Account -> #12 (p4); Account -> #13 (p5); Account -> #14 (p5); Account -> #15 (p5); Account -> #16 (p5); Account -> #17 (p6); Account -> #18 (p6)

#02 / Frontend Action / Implemented

Bags / luggage catalog

Catchall storefront resolves bags, luggage, categories and collection routes.

src/app/[...segments]/page.tsx:207

Next -> #3 (p2)

#03 / Frontend Action / Implemented

Listings and product details

Catalog facets and /products/[slug] detail.

src/components/catalog-view.tsx:1

Next -> #4 (p2)

#04 / Frontend Action / Implemented

Cart and checkout

Store provider and cart page send server cart / checkout requests.

src/components/store-provider.tsx:1

Next -> #5 (p3)

Frontend architecture

Readable detail sheet 2/5. Node numbers and page references preserve every directed connection.

#05 / HTTP Request / Implemented

Storefront API requests

Actual client/server routes documented in the route inventory.

src/app/api/v1/[[...path]]/route.ts:42

Next -> #6 (p3); Failure -> #8 (p3)

#06 / Backend Service / Implemented

Server processing

Services return current catalog, bag, quote and order data.

src/server/commerce.ts:50

Next -> #7 (p3)

#07 / Response / Implemented

Render confirmed result

State updates follow API results, with loading and error UI.

src/components/store-provider.tsx:1

END: no outgoing connection

#08 / Error Handler / Implemented

Request failed

Validation, access, conflict or service errors are returned to the caller. Status and error shape depend on the specific route.

src/server/http.ts:69

Next -> #9 (p4)

Frontend architecture

Readable detail sheet 3/5. Node numbers and page references preserve every directed connection.

#09 / Response / Implemented

Show error / retry

Failure end state. Caller receives the actual error; a retry may be offered by the UI.

src/server/http.ts:69

END: no outgoing connection

#10 / Frontend Action / Implemented

Registration / OTP

Customer account section backed by actual account, care, order or review APIs. Address book is a separate component.

src/components/auth-experience.tsx:1

Next -> #5 (p3)

#11 / Frontend Action / Implemented

Google / Apple sign-in

Customer account section backed by actual account, care, order or review APIs. Address book is a separate component.

src/components/social-sign-in-buttons.tsx:1

Next -> #5 (p3)

#12 / Frontend Action / Implemented

Account dashboard

Customer account section backed by actual account, care, order or review APIs. Address book is a separate component.

src/components/account-dashboard.tsx:1

Next -> #5 (p3)

Frontend architecture

Readable detail sheet 4/5. Node numbers and page references preserve every directed connection.

#13 / Frontend Action / Implemented

Profile and addresses

Customer account section backed by actual account, care, order or review APIs. Address book is a separate component.

src/components/customer-profile.tsx:1

Next -> #5 (p3)

#14 / Frontend Action / Implemented

Orders and tracking

Customer account section backed by actual account, care, order or review APIs. Address book is a separate component.

src/components/account-orders-portal.tsx:1

Next -> #5 (p3)

#15 / Frontend Action / Implemented

Cancellation / returns

Customer account section backed by actual account, care, order or review APIs. Address book is a separate component.

src/components/account-orders-returns.tsx:1

Next -> #5 (p3)

#16 / Frontend Action / Implemented

Refund / notifications

Customer account section backed by actual account, care, order or review APIs. Address book is a separate component.

src/components/account-activity-feed.tsx:1

Next -> #5 (p3)

Frontend architecture

Readable detail sheet 5/5. Node numbers and page references preserve every directed connection.

#17 / Frontend Action / Implemented

Product reviews

Customer account section backed by actual account, care, order or review APIs. Address book is a separate component.

src/components/account-reviews.tsx:1

Next -> #5 (p3)

#18 / Frontend Action / Implemented

Customer support

Customer account section backed by actual account, care, order or review APIs. Address book is a separate component.

src/components/account-support.tsx:1

Next -> #5 (p3)

Frontend architecture

Complete directed connection register / no omitted success, failure or proposed edges

#01 Home and navigation -> #02 Bags / luggage catalog | Next

#02 Bags / luggage catalog -> #03 Listings and product details | Next

#03 Listings and product details -> #04 Cart and checkout | Next

#04 Cart and checkout -> #05 Storefront API requests | Next

#05 Storefront API requests -> #06 Server processing | Next

#06 Server processing -> #07 Render confirmed result | Next

#05 Storefront API requests -> #08 Request failed | Failure | FAILURE

#08 Request failed -> #09 Show error / retry | Next | FAILURE

#01 Home and navigation -> #10 Registration / OTP | Account

#10 Registration / OTP -> #05 Storefront API requests | Next

#01 Home and navigation -> #11 Google / Apple sign-in | Account

#11 Google / Apple sign-in -> #05 Storefront API requests | Next

#01 Home and navigation -> #12 Account dashboard | Account

#12 Account dashboard -> #05 Storefront API requests | Next

#01 Home and navigation -> #13 Profile and addresses | Account

#13 Profile and addresses -> #05 Storefront API requests | Next

#01 Home and navigation -> #14 Orders and tracking | Account

#14 Orders and tracking -> #05 Storefront API requests | Next

#01 Home and navigation -> #15 Cancellation / returns | Account

#15 Cancellation / returns -> #05 Storefront API requests | Next

#01 Home and navigation -> #16 Refund / notifications | Account

#16 Refund / notifications -> #05 Storefront API requests | Next

Frontend architecture

Directed connection register continued

#01 Home and navigation -> #17 Product reviews | Account

#17 Product reviews -> #05 Storefront API requests | Next

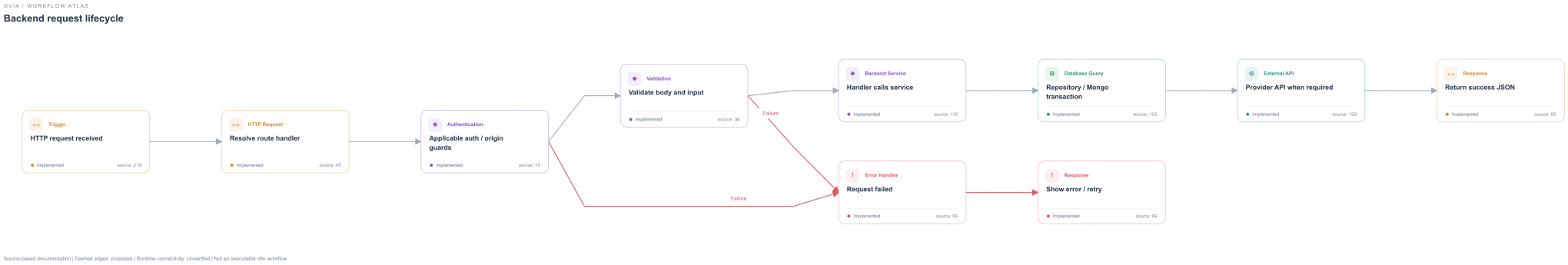
#01 Home and navigation -> #18 Customer support | Account

#18 Customer support -> #05 Storefront API requests | Next

Note: Source implementation is documented; deployed configuration and provider connectivity are not verified.

Backend request lifecycle

Architecture overview. Readable node and connection sheets follow.



Backend request lifecycle

Readable detail sheet 1/3. Node numbers and page references preserve every directed connection.

#01 / Trigger / Implemented

HTTP request received

Next.js route receives request and promised path params.

src/app/api/v1/[[...path]]/route.ts:219

Next -> #2 (p2)

#02 / HTTP Request / Implemented

Resolve route handler

App Router file routes plus explicit catchall dispatch.

src/app/api/v1/[[...path]]/route.ts:45

Next -> #3 (p2)

#03 / Authentication / Implemented

Applicable auth / origin guards

Each handler decides which helpers apply; this is a representative guarded mutation lifecycle.

src/server/http.ts:15

Next -> #4 (p2); Failure -> #9 (p4)

#04 / Validation / Implemented

Validate body and input

Streaming body cap and Zod validation where explicitly called.

src/server/http.ts:36

Next -> #5 (p3); Failure -> #9 (p4)

Backend request lifecycle

Readable detail sheet 2/3. Node numbers and page references preserve every directed connection.

#05 / Backend Service / Implemented

Handler calls service

Actual business rules and idempotency are in server modules.

src/server/commerce.ts:115

Next -> #6 (p3)

#06 / Database Query / Implemented

Repository / Mongo transaction

Native database operations share a session when required.

src/server/db.ts:103

Next -> #7 (p3)

#07 / External API / Implemented

Provider API when required

External writes are reconciled with durable local state; not all requests call a provider. Runtime configuration and connectivity are Unverified.

src/server/razorpay-checkout.ts:109

Next -> #8 (p3)

#08 / Response / Implemented

Return success JSON

Common wrapper returns {data}; other routes use documented custom response shapes.

src/server/http.ts:69

END: no outgoing connection

Backend request lifecycle

Readable detail sheet 3/3. Node numbers and page references preserve every directed connection.

#09 / Error Handler / Implemented

Request failed

Validation, access, conflict or service errors are returned to the caller. Status and error shape depend on the specific route.

src/server/http.ts:69

Next -> #10 (p4)

#10 / Response / Implemented

Show error / retry

Failure end state. Caller receives the actual error; a retry may be offered by the UI.

src/server/http.ts:69

END: no outgoing connection

Backend request lifecycle

Complete directed connection register / no omitted success, failure or proposed edges

#01 HTTP request received -> #02 Resolve route handler | Next

#02 Resolve route handler -> #03 Applicable auth / origin guards | Next

#03 Applicable auth / origin guards -> #04 Validate body and input | Next

#04 Validate body and input -> #05 Handler calls service | Next

#05 Handler calls service -> #06 Repository / Mongo transaction | Next

#06 Repository / Mongo transaction -> #07 Provider API when required | Next

#07 Provider API when required -> #08 Return success JSON | Next

#03 Applicable auth / origin guards -> #09 Request failed | Failure | FAILURE

#04 Validate body and input -> #09 Request failed | Failure | FAILURE

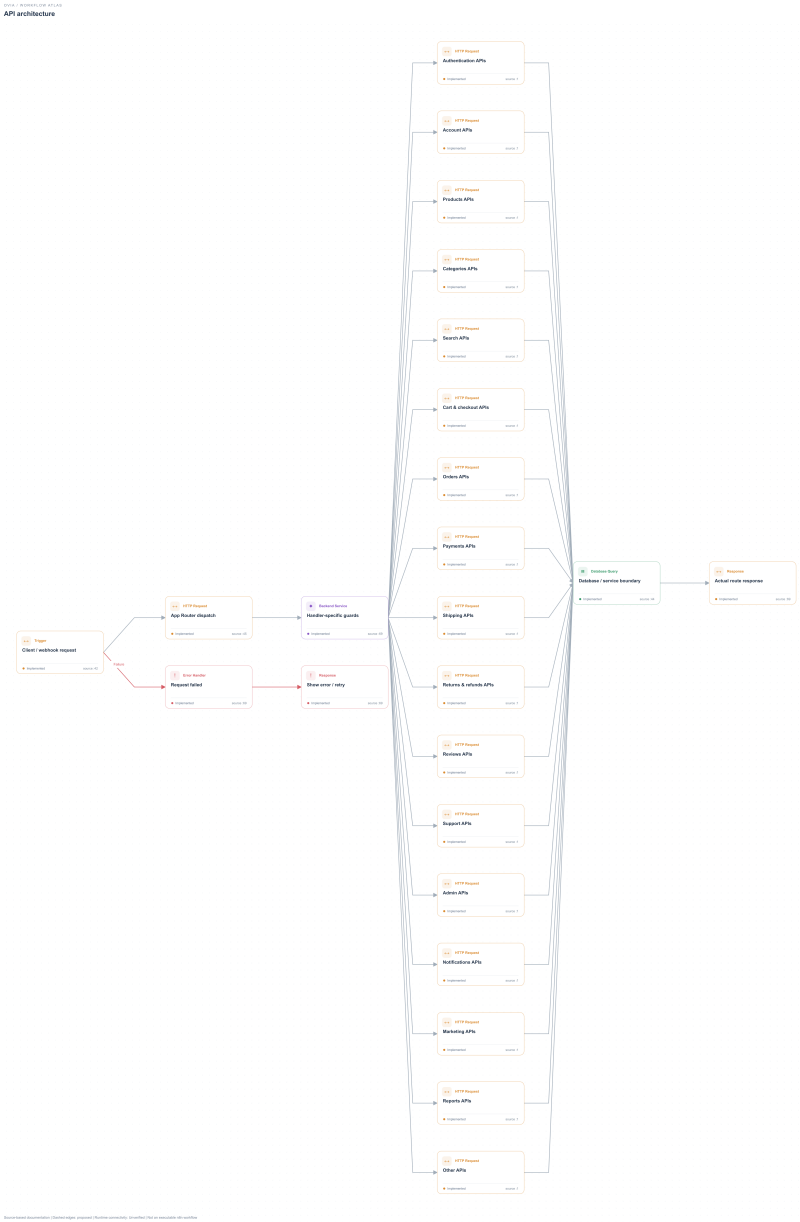
#09 Request failed -> #10 Show error / retry | Next | FAILURE

Note: Source implementation is documented; deployed configuration and provider connectivity are not verified.

Note: Source contains no central src/middleware.ts or src/proxy.ts. Guards are route-local and server-component-local. Logging is selective, not a universal centralized logging pipeline.

API architecture

Architecture overview. Readable node and connection sheets follow.



API architecture

Readable detail sheet 1/6. Node numbers and page references preserve every directed connection.

#01 / Trigger / Implemented

Client / webhook request

Customer, owner, cron and provider entry points.

src/app/api/v1/[[...path]]/route.ts:42

Next -> #2 (p2); Failure -> #6 (p3)

#02 / HTTP Request / Implemented

App Router dispatch

Inventory records every exported method from all 86 route files; catchalls have an additional explicit branch inventory.

src/app/api/v1/[[...path]]/route.ts:45

Next -> #3 (p2)

#03 / Backend Service / Implemented

Handler-specific guards

Auth, origin, rate limits, validation and response conventions vary by endpoint.

src/server/http.ts:69

Next -> #8 (p3); Next -> #9 (p4); Next -> #10 (p4); Next -> #11 (p4); Next -> #12 (p4); Next -> #13 (p5); Next -> #14 (p5); Next -> #15 (p5); Next -> #16 (p5); Next -> #17 (p6); Next -> #18 (p6); Next -> #19 (p6); Next -> #20 (p6); Next -> #21 (p7);...

#04 / Database Query / Implemented

Database / service boundary

Imported service symbols and source file evidence are retained per operation.

src/server/db.ts:44

Next -> #5 (p3)

API architecture

Readable detail sheet 2/6. Node numbers and page references preserve every directed connection.

#05 / Response / Implemented

Actual route response

Use the inventory and per-operation graph to inspect the full source-derived lifecycle.

src/server/http.ts:69

END: no outgoing connection

#06 / Error Handler / Implemented

Request failed

Validation, access, conflict or service errors are returned to the caller. Status and error shape depend on the specific route.

src/server/http.ts:69

Next -> #7 (p3)

#07 / Response / Implemented

Show error / retry

Failure end state. Caller receives the actual error; a retry may be offered by the UI.

src/server/http.ts:69

END: no outgoing connection

#08 / HTTP Request / Implemented

Authentication APIs

5 exported file-route operations. See inventory for individual handlers, service symbols and detected guards.

src/app/api/admin/auth/login/route.ts:1

Next -> #4 (p2)

API architecture

Readable detail sheet 3/6. Node numbers and page references preserve every directed connection.

#09 / HTTP Request / Implemented

Account APIs

14 exported file-route operations. See inventory for individual handlers, service symbols and detected guards.

src/app/api/v1/account/activity/route.ts:1

Next -> #4 (p2)

#10 / HTTP Request / Implemented

Products APIs

2 exported file-route operations. See inventory for individual handlers, service symbols and detected guards.

src/app/api/products/route.ts:1

Next -> #4 (p2)

#11 / HTTP Request / Implemented

Categories APIs

1 exported file-route operations. See inventory for individual handlers, service symbols and detected guards.

src/app/api/categories/route.ts:1

Next -> #4 (p2)

#12 / HTTP Request / Implemented

Search APIs

1 exported file-route operations. See inventory for individual handlers, service symbols and detected guards.

src/app/api/search/route.ts:1

Next -> #4 (p2)

API architecture

Readable detail sheet 4/6. Node numbers and page references preserve every directed connection.

#13 / HTTP Request / Implemented

Cart & checkout APIs

2 exported file-route operations. See inventory for individual handlers, service symbols and detected guards.

src/app/api/v1/checkout/[[...action]]/route.ts:1

Next -> #4 (p2)

#14 / HTTP Request / Implemented

Orders APIs

4 exported file-route operations. See inventory for individual handlers, service symbols and detected guards.

src/app/api/v1/bulk-orders/route.ts:1

Next -> #4 (p2)

#15 / HTTP Request / Implemented

Payments APIs

2 exported file-route operations. See inventory for individual handlers, service symbols and detected guards.

src/app/api/v1/payments/razorpay/[[...action]]/route.ts:1

Next -> #4 (p2)

#16 / HTTP Request / Implemented

Shipping APIs

3 exported file-route operations. See inventory for individual handlers, service symbols and detected guards.

src/app/api/orders/[orderId]/tracking/route.ts:1

Next -> #4 (p2)

API architecture

Readable detail sheet 5/6. Node numbers and page references preserve every directed connection.

#17 / HTTP Request / Implemented

Returns & refunds APIs

1 exported file-route operations. See inventory for individual handlers, service symbols and detected guards.

src/app/api/v1/orders/[id]/return/route.ts:1

Next -> #4 (p2)

#18 / HTTP Request / Implemented

Reviews APIs

4 exported file-route operations. See inventory for individual handlers, service symbols and detected guards.

src/app/api/v1/reviews/route.ts:1

Next -> #4 (p2)

#19 / HTTP Request / Implemented

Support APIs

11 exported file-route operations. See inventory for individual handlers, service symbols and detected guards.

src/app/api/internal/support/sync/route.ts:1

Next -> #4 (p2)

#20 / HTTP Request / Implemented

Admin APIs

70 exported file-route operations. See inventory for individual handlers, service symbols and detected guards.

src/app/api/admin/abandoned-carts/route.ts:1

Next -> #4 (p2)

API architecture

Readable detail sheet 6/6. Node numbers and page references preserve every directed connection.

#21 / HTTP Request / Implemented

Notifications APIs

3 exported file-route operations. See inventory for individual handlers, service symbols and detected guards.

src/app/api/admin/notifications/route.ts:1

Next -> #4 (p2)

#22 / HTTP Request / Implemented

Marketing APIs

13 exported file-route operations. See inventory for individual handlers, service symbols and detected guards.

src/app/api/admin/coupons/route.ts:1

Next -> #4 (p2)

#23 / HTTP Request / Implemented

Reports APIs

2 exported file-route operations. See inventory for individual handlers, service symbols and detected guards.

src/app/api/admin/dashboard/route.ts:1

Next -> #4 (p2)

#24 / HTTP Request / Implemented

Other APIs

14 exported file-route operations. See inventory for individual handlers, service symbols and detected guards.

src/app/api/cron/expire/route.ts:1

Next -> #4 (p2)

API architecture

Complete directed connection register / no omitted success, failure or proposed edges

#01 Client / webhook request -> #02 App Router dispatch | Next

#02 App Router dispatch -> #03 Handler-specific guards | Next

#04 Database / service boundary -> #05 Actual route response | Next

#01 Client / webhook request -> #06 Request failed | Failure | FAILURE

#06 Request failed -> #07 Show error / retry | Next | FAILURE

#03 Handler-specific guards -> #08 Authentication APIs | Next

#08 Authentication APIs -> #04 Database / service boundary | Next

#03 Handler-specific guards -> #09 Account APIs | Next

#09 Account APIs -> #04 Database / service boundary | Next

#03 Handler-specific guards -> #10 Products APIs | Next

#10 Products APIs -> #04 Database / service boundary | Next

#03 Handler-specific guards -> #11 Categories APIs | Next

#11 Categories APIs -> #04 Database / service boundary | Next

#03 Handler-specific guards -> #12 Search APIs | Next

#12 Search APIs -> #04 Database / service boundary | Next

#03 Handler-specific guards -> #13 Cart & checkout APIs | Next

#13 Cart & checkout APIs -> #04 Database / service boundary | Next

#03 Handler-specific guards -> #14 Orders APIs | Next

#14 Orders APIs -> #04 Database / service boundary | Next

#03 Handler-specific guards -> #15 Payments APIs | Next

#15 Payments APIs -> #04 Database / service boundary | Next

#03 Handler-specific guards -> #16 Shipping APIs | Next

API architecture

Directed connection register continued

#16 Shipping APIs -> #04 Database / service boundary | Next

#03 Handler-specific guards -> #17 Returns & refunds APIs | Next

#17 Returns & refunds APIs -> #04 Database / service boundary | Next

#03 Handler-specific guards -> #18 Reviews APIs | Next

#18 Reviews APIs -> #04 Database / service boundary | Next

#03 Handler-specific guards -> #19 Support APIs | Next

#19 Support APIs -> #04 Database / service boundary | Next

#03 Handler-specific guards -> #20 Admin APIs | Next

#20 Admin APIs -> #04 Database / service boundary | Next

#03 Handler-specific guards -> #21 Notifications APIs | Next

#21 Notifications APIs -> #04 Database / service boundary | Next

#03 Handler-specific guards -> #22 Marketing APIs | Next

#22 Marketing APIs -> #04 Database / service boundary | Next

#03 Handler-specific guards -> #23 Reports APIs | Next

#23 Reports APIs -> #04 Database / service boundary | Next

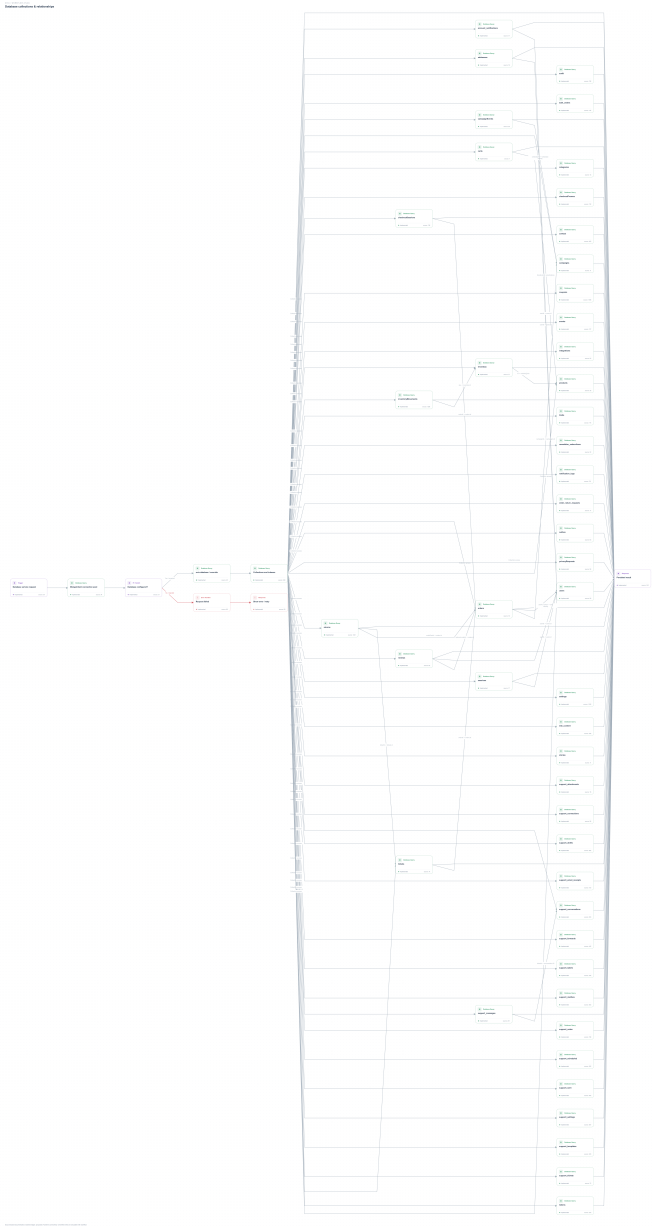
#03 Handler-specific guards -> #24 Other APIs | Next

#24 Other APIs -> #04 Database / service boundary | Next

Note: Source implementation is documented; deployed configuration and provider connectivity are not verified.

Database collections & relationships

Architecture overview. Readable node and connection sheets follow.



Database collections & relationships

Readable detail sheet 1/14. Node numbers and page references preserve every directed connection.

#01 / Trigger / Implemented

Database service request

Backend requests a shared MongoDB client.

src/server/db.ts:30

Next -> #2 (p2)

#02 / Database Query / Implemented

MongoClient connection pool

MONGODB_URI environment lookup, pool of 10 and six-second selection timeout. No URI is displayed.

src/server/db.ts:34

Next -> #3 (p2)

#03 / IF / Switch / Implemented

Database configured?

Live services fail without MongoDB. A legacy local adapter exists after an unconditional early rejection and is not a working fallback.

src/server/db.ts:31

Yes / accepted -> #4 (p2); No / rejected -> #7 (p3)

#04 / Database Query / Implemented

ovia database / override

Configured DB name or ovia default; deployed Atlas topology is Unverified.

src/server/db.ts:45

Next -> #5 (p3)

Database collections & relationships

Readable detail sheet 2/14. Node numbers and page references preserve every directed connection.

#05 / Database Query / Implemented

Collections and indexes

setupDatabase defines identity, email uniqueness, token TTL and checkout idempotency indexes.

src/server/db.ts:239

Next -> #6 (p3); Collection access -> #9 (p4); Collection access -> #10 (p4); Collection access -> #11 (p4); Collection access -> #12 (p4); Collection access -> #13 (p5); Collection access -> #14 (p5); Collection access -> #15 (p5); Collection...

#06 / Response / Implemented

Persisted result

Repository reads and transactional writes return stored records.

src/server/db.ts:100

END: no outgoing connection

#07 / Error Handler / Implemented

Request failed

Validation, access, conflict or service errors are returned to the caller. Status and error shape depend on the specific route.

src/server/http.ts:69

Next -> #8 (p3)

#08 / Response / Implemented

Show error / retry

Failure end state. Caller receives the actual error; a retry may be offered by the UI.

src/server/http.ts:69

END: no outgoing connection

Database collections & relationships

Readable detail sheet 3/14. Node numbers and page references preserve every directed connection.

#09 / Database Query / Implemented

account_notifications

Actual collection name used by source. Evidence: 7 literal references. Production contents are Unverified.

src/app/api/v1/account/activity/route.ts:17

Next -> #6 (p3); userId -> users.id -> #56 (p15)

#10 / Database Query / Implemented

addresses

Actual collection name used by source. Evidence: 9 literal references. Production contents are Unverified.

src/app/api/v1/account/addresses/[[...id]]/route.ts:19

Next -> #6 (p3); userId -> users.id -> #56 (p15)

#11 / Database Query / Implemented

audit

Actual collection name used by source. Evidence: 13 literal references. Production contents are Unverified.

src/app/api/admin/support/route.ts:196

Next -> #6 (p3)

#12 / Database Query / Implemented

bulk_orders

Actual collection name used by source. Evidence: 3 literal references. Production contents are Unverified.

src/app/api/v1/bulk-orders/route.ts:142

Next -> #6 (p3)

Database collections & relationships

Readable detail sheet 4/14. Node numbers and page references preserve every directed connection.

#13 / Database Query / Implemented

campaignEvents

Actual collection name used by source. Evidence: 3 literal references. Production contents are Unverified.

src/server/campaigns.ts:20

Next -> #6 (p3); campaignId -> campaign identity -> #14 (p5)

#14 / Database Query / Implemented

campaigns

Actual collection name used by source. Evidence: 4 literal references. Production contents are Unverified.

src/server/campaigns.ts:11

Next -> #6 (p3)

#15 / Database Query / Implemented

carts

Actual collection name used by source. Evidence: 18 literal references. Production contents are Unverified.

src/server/account-sign-in.ts:7

Next -> #6 (p3); lines[].sku -> variants[].sku -> #32 (p9)

#16 / Database Query / Implemented

categories

Actual collection name used by source. Evidence: 5 literal references. Production contents are Unverified.

src/app/api/admin/product-taxonomy/route.ts:14

Next -> #6 (p3)

Database collections & relationships

Readable detail sheet 5/14. Node numbers and page references preserve every directed connection.

#17 / Database Query / Implemented

checkoutPromos

Actual collection name used by source. Evidence: 3 literal references. Production contents are Unverified.

src/server/checkout-pricing.ts:132

Next -> #6 (p3)

#18 / Database Query / Implemented

checkoutSessions

Actual collection name used by source. Evidence: 15 literal references. Production contents are Unverified.

src/server/commerce.ts:136

Next -> #6 (p3); orderId -> orders.id -> #29 (p9)

#19 / Database Query / Implemented

contact

Actual collection name used by source. Evidence: 4 literal references. Production contents are Unverified.

src/app/api/v1/[[...path]]/route.ts:450

Next -> #6 (p3)

#20 / Database Query / Implemented

coupons

Actual collection name used by source. Evidence: 18 literal references. Production contents are Unverified.

src/server/admin-service.ts:1546

Next -> #6 (p3)

Database collections & relationships

Readable detail sheet 6/14. Node numbers and page references preserve every directed connection.

#21 / Database Query / Implemented

events

Actual collection name used by source. Evidence: 20 literal references. Production contents are Unverified.

src/app/api/admin/content/blog-events/route.ts:127

Next -> #6 (p3)

#22 / Database Query / Implemented

integrations

Actual collection name used by source. Evidence: 6 literal references. Production contents are Unverified.

src/server/integrations-service.ts:53

Next -> #6 (p3)

#23 / Database Query / Implemented

inventory

Actual collection name used by source. Evidence: 17 literal references. Production contents are Unverified.

src/app/api/v1/orders/[id]/return/route.ts:77

Next -> #6 (p3); id -> variants[].sku -> #32 (p9)

#24 / Database Query / Implemented

inventoryMovements

Actual collection name used by source. Evidence: 2 literal references. Production contents are Unverified.

src/server/admin-service.ts:1206

Next -> #6 (p3); sku -> inventory.id -> #23 (p7)

Database collections & relationships

Readable detail sheet 7/14. Node numbers and page references preserve every directed connection.

#25 / Database Query / Implemented

limits

Actual collection name used by source. Evidence: 5 literal references. Production contents are Unverified.

src/server/http.ts:114

Next -> #6 (p3)

#26 / Database Query / Implemented

newsletter_subscribers

Actual collection name used by source. Evidence: 1 literal references. Production contents are Unverified.

src/app/api/v1/account/newsletter/route.ts:22

Next -> #6 (p3)

#27 / Database Query / Implemented

notification_logs

Actual collection name used by source. Evidence: 6 literal references. Production contents are Unverified.

src/server/notification-service.ts:751

Next -> #6 (p3)

#28 / Database Query / Implemented

order_return_requests

Actual collection name used by source. Evidence: 3 literal references. Production contents are Unverified.

src/app/api/admin/orders/[id]/return-request/route.ts:17

Next -> #6 (p3)

Database collections & relationships

Readable detail sheet 8/14. Node numbers and page references preserve every directed connection.

#29 / Database Query / Implemented

orders

Actual collection name used by source. Evidence: 112 literal references. Production contents are Unverified.

src/app/api/admin/orders/[id]/document/route.ts:16

Next -> #6 (p3); userId / owner -> user identity -> #56 (p15); campaignId -> campaigns.id -> #14 (p5)

#30 / Database Query / Implemented

outbox

Actual collection name used by source. Evidence: 30 literal references. Production contents are Unverified.

src/app/api/admin/shipping/route.ts:63

Next -> #6 (p3)

#31 / Database Query / Implemented

privacyRequests

Actual collection name used by source. Evidence: 8 literal references. Production contents are Unverified.

src/app/api/v1/care/[[...path]]/route.ts:58

Next -> #6 (p3)

#32 / Database Query / Implemented

products

Actual collection name used by source. Evidence: 20 literal references. Production contents are Unverified.

src/app/api/admin/product-taxonomy/route.ts:30

Next -> #6 (p3)

Database collections & relationships

Readable detail sheet 9/14. Node numbers and page references preserve every directed connection.

#33 / Database Query / Implemented

returns

Actual collection name used by source. Evidence: 2 literal references. Production contents are Unverified.

src/server/admin-service.ts:1401

Next -> #6 (p3); ticketId -> tickets.id -> #54 (p15); orderRawId -> orders.id -> #29 (p9)

#34 / Database Query / Implemented

reviews

Actual collection name used by source. Evidence: 9 literal references. Production contents are Unverified.

src/app/api/v1/reviews/route.ts:63

Next -> #6 (p3); userId -> users.id -> #56 (p15); optional orderId -> orders.id -> #29 (p9)

#35 / Database Query / Implemented

sessions

Actual collection name used by source. Evidence: 24 literal references. Production contents are Unverified.

src/app/api/admin/security/route.ts:17

Next -> #6 (p3); userId -> users.id -> #56 (p15)

#36 / Database Query / Implemented

settings

Actual collection name used by source. Evidence: 4 literal references. Production contents are Unverified.

src/server/admin-service.ts:1818

Next -> #6 (p3)

Database collections & relationships

Readable detail sheet 10/14. Node numbers and page references preserve every directed connection.

#37 / Database Query / Implemented

site_content

Actual collection name used by source. Evidence: 4 literal references. Production contents are Unverified.

src/app/api/admin/content/route.ts:230

Next -> #6 (p3)

#38 / Database Query / Implemented

stories

Actual collection name used by source. Evidence: 16 literal references. Production contents are Unverified.

src/app/api/admin/articles/route.ts:11

Next -> #6 (p3)

#39 / Database Query / Implemented

support_attachments

Actual collection name used by source. Evidence: 6 literal references. Production contents are Unverified.

src/app/api/support/attachments/[id]/route.ts:18

Next -> #6 (p3)

#40 / Database Query / Implemented

support_connections

Actual collection name used by source. Evidence: 5 literal references. Production contents are Unverified.

src/app/api/admin/support/gmail/callback/route.ts:58

Next -> #6 (p3)

Database collections & relationships

Readable detail sheet 11/14. Node numbers and page references preserve every directed connection.

#41 / Database Query / Implemented

support_conversations

Actual collection name used by source. Evidence: 17 literal references. Production contents are Unverified.

src/app/api/v1/care/[[...path]]/route.ts:453

Next -> #6 (p3)

#42 / Database Query / Implemented

support_drafts

Actual collection name used by source. Evidence: 4 literal references. Production contents are Unverified.

src/server/support-mailbox.ts:283

Next -> #6 (p3)

#43 / Database Query / Implemented

support_email_receipts

Actual collection name used by source. Evidence: 7 literal references. Production contents are Unverified.

src/server/support-gmail.ts:153

Next -> #6 (p3)

#44 / Database Query / Implemented

support_forwards

Actual collection name used by source. Evidence: 3 literal references. Production contents are Unverified.

src/server/support-mailbox.ts:442

Next -> #6 (p3)

Database collections & relationships

Readable detail sheet 12/14. Node numbers and page references preserve every directed connection.

#45 / Database Query / Implemented

support_labels

Actual collection name used by source. Evidence: 3 literal references. Production contents are Unverified.

src/server/support-mailbox.ts:343

Next -> #6 (p3)

#46 / Database Query / Implemented

support_mailbox

Actual collection name used by source. Evidence: 8 literal references. Production contents are Unverified.

src/server/support-inbox.ts:363

Next -> #6 (p3)

#47 / Database Query / Implemented

support_messages

Actual collection name used by source. Evidence: 7 literal references. Production contents are Unverified.

src/app/api/v1/care/[[...path]]/route.ts:231

Next -> #6 (p3); ticketId -> conversation ID -> #41 (p12)

#48 / Database Query / Implemented

support_notes

Actual collection name used by source. Evidence: 5 literal references. Production contents are Unverified.

src/server/support-conversations.ts:198

Next -> #6 (p3)

Database collections & relationships

Readable detail sheet 13/14. Node numbers and page references preserve every directed connection.

#49 / Database Query / Implemented

support_scheduled

Actual collection name used by source. Evidence: 6 literal references. Production contents are Unverified.

src/server/support-mailbox.ts:305

Next -> #6 (p3)

#50 / Database Query / Implemented

support_sent

Actual collection name used by source. Evidence: 4 literal references. Production contents are Unverified.

src/server/support-mailbox.ts:403

Next -> #6 (p3)

#51 / Database Query / Implemented

support_settings

Actual collection name used by source. Evidence: 2 literal references. Production contents are Unverified.

src/server/support-mailbox.ts:367

Next -> #6 (p3)

#52 / Database Query / Implemented

support_templates

Actual collection name used by source. Evidence: 3 literal references. Production contents are Unverified.

src/server/support-mailbox.ts:323

Next -> #6 (p3)

Database collections & relationships

Readable detail sheet 14/14. Node numbers and page references preserve every directed connection.

#53 / Database Query / Implemented

support_tickets

Actual collection name used by source. Evidence: 7 literal references. Production contents are Unverified.

src/app/api/v1/support/route.ts:75

Next -> #6 (p3)

#54 / Database Query / Implemented

tickets

Actual collection name used by source. Evidence: 18 literal references. Production contents are Unverified.

src/app/api/v1/account/activity/route.ts:14

Next -> #6 (p3); orderId -> orders.id -> #29 (p9)

#55 / Database Query / Implemented

tokens

Actual collection name used by source. Evidence: 29 literal references. Production contents are Unverified.

src/server/db.ts:270

Next -> #6 (p3)

#56 / Database Query / Implemented

users

Actual collection name used by source. Evidence: 53 literal references. Production contents are Unverified.

src/app/api/admin/customers/route.ts:59

Next -> #6 (p3)

Database collections & relationships

Complete directed connection register / no omitted success, failure or proposed edges

#01 Database service request -> #02 MongoClient connection pool | Next

#02 MongoClient connection pool -> #03 Database configured? | Next

#03 Database configured? -> #04 ovia database / override | Yes / accepted

#04 ovia database / override -> #05 Collections and indexes | Next

#05 Collections and indexes -> #06 Persisted result | Next

#03 Database configured? -> #07 Request failed | No / rejected | FAILURE

#07 Request failed -> #08 Show error / retry | Next | FAILURE

#05 Collections and indexes -> #09 account_notifications | Collection access

#09 account_notifications -> #06 Persisted result | Next

#05 Collections and indexes -> #10 addresses | Collection access

#10 addresses -> #06 Persisted result | Next

#05 Collections and indexes -> #11 audit | Collection access

#11 audit -> #06 Persisted result | Next

#05 Collections and indexes -> #12 bulk_orders | Collection access

#12 bulk_orders -> #06 Persisted result | Next

#05 Collections and indexes -> #13 campaignEvents | Collection access

#13 campaignEvents -> #06 Persisted result | Next

#05 Collections and indexes -> #14 campaigns | Collection access

#14 campaigns -> #06 Persisted result | Next

#05 Collections and indexes -> #15 carts | Collection access

#15 carts -> #06 Persisted result | Next

#05 Collections and indexes -> #16 categories | Collection access

Database collections & relationships

Directed connection register continued

#16 categories -> #06 Persisted result | Next

#05 Collections and indexes -> #17 checkoutPromos | Collection access

#17 checkoutPromos -> #06 Persisted result | Next

#05 Collections and indexes -> #18 checkoutSessions | Collection access

#18 checkoutSessions -> #06 Persisted result | Next

#05 Collections and indexes -> #19 contact | Collection access

#19 contact -> #06 Persisted result | Next

#05 Collections and indexes -> #20 coupons | Collection access

#20 coupons -> #06 Persisted result | Next

#05 Collections and indexes -> #21 events | Collection access

#21 events -> #06 Persisted result | Next

#05 Collections and indexes -> #22 integrations | Collection access

#22 integrations -> #06 Persisted result | Next

#05 Collections and indexes -> #23 inventory | Collection access

#23 inventory -> #06 Persisted result | Next

#05 Collections and indexes -> #24 inventoryMovements | Collection access

#24 inventoryMovements -> #06 Persisted result | Next

#05 Collections and indexes -> #25 limits | Collection access

#25 limits -> #06 Persisted result | Next

#05 Collections and indexes -> #26 newsletter_subscribers | Collection access

#26 newsletter_subscribers -> #06 Persisted result | Next

#05 Collections and indexes -> #27 notification_logs | Collection access

Database collections & relationships

Directed connection register continued

#27 notification_logs -> #06 Persisted result | Next

#05 Collections and indexes -> #28 order_return_requests | Collection access

#28 order_return_requests -> #06 Persisted result | Next

#05 Collections and indexes -> #29 orders | Collection access

#29 orders -> #06 Persisted result | Next

#05 Collections and indexes -> #30 outbox | Collection access

#30 outbox -> #06 Persisted result | Next

#05 Collections and indexes -> #31 privacyRequests | Collection access

#31 privacyRequests -> #06 Persisted result | Next

#05 Collections and indexes -> #32 products | Collection access

#32 products -> #06 Persisted result | Next

#05 Collections and indexes -> #33 returns | Collection access

#33 returns -> #06 Persisted result | Next

#05 Collections and indexes -> #34 reviews | Collection access

#34 reviews -> #06 Persisted result | Next

#05 Collections and indexes -> #35 sessions | Collection access

#35 sessions -> #06 Persisted result | Next

#05 Collections and indexes -> #36 settings | Collection access

#36 settings -> #06 Persisted result | Next

#05 Collections and indexes -> #37 site_content | Collection access

#37 site_content -> #06 Persisted result | Next

#05 Collections and indexes -> #38 stories | Collection access

Database collections & relationships

Directed connection register continued

#38 stories -> #06 Persisted result | Next

#05 Collections and indexes -> #39 support_attachments | Collection access

#39 support_attachments -> #06 Persisted result | Next

#05 Collections and indexes -> #40 support_connections | Collection access

#40 support_connections -> #06 Persisted result | Next

#05 Collections and indexes -> #41 support_conversations | Collection access

#41 support_conversations -> #06 Persisted result | Next

#05 Collections and indexes -> #42 support_drafts | Collection access

#42 support_drafts -> #06 Persisted result | Next

#05 Collections and indexes -> #43 support_email_receipts | Collection access

#43 support_email_receipts -> #06 Persisted result | Next

#05 Collections and indexes -> #44 support_forwards | Collection access

#44 support_forwards -> #06 Persisted result | Next

#05 Collections and indexes -> #45 support_labels | Collection access

#45 support_labels -> #06 Persisted result | Next

#05 Collections and indexes -> #46 support_mailbox | Collection access

#46 support_mailbox -> #06 Persisted result | Next

#05 Collections and indexes -> #47 support_messages | Collection access

#47 support_messages -> #06 Persisted result | Next

#05 Collections and indexes -> #48 support_notes | Collection access

#48 support_notes -> #06 Persisted result | Next

#05 Collections and indexes -> #49 support_scheduled | Collection access

Database collections & relationships

Directed connection register continued

#49 support_scheduled -> #06 Persisted result | Next

#05 Collections and indexes -> #50 support_sent | Collection access

#50 support_sent -> #06 Persisted result | Next

#05 Collections and indexes -> #51 support_settings | Collection access

#51 support_settings -> #06 Persisted result | Next

#05 Collections and indexes -> #52 support_templates | Collection access

#52 support_templates -> #06 Persisted result | Next

#05 Collections and indexes -> #53 support_tickets | Collection access

#53 support_tickets -> #06 Persisted result | Next

#05 Collections and indexes -> #54 tickets | Collection access

#54 tickets -> #06 Persisted result | Next

#05 Collections and indexes -> #55 tokens | Collection access

#55 tokens -> #06 Persisted result | Next

#05 Collections and indexes -> #56 users | Collection access

#56 users -> #06 Persisted result | Next

#35 sessions -> #56 users | userId -> users.id

#10 addresses -> #56 users | userId -> users.id

#29 orders -> #56 users | userId / owner -> user identity

#18 checkoutSessions -> #29 orders | orderId -> orders.id

#23 inventory -> #32 products | id -> variants[].sku

#15 carts -> #32 products | lines[].sku -> variants[].sku

#34 reviews -> #56 users | userId -> users.id

Database collections & relationships

Directed connection register continued

#34 reviews -> #29 orders | optional orderId -> orders.id

#54 tickets -> #29 orders | orderId -> orders.id

#33 returns -> #54 tickets | ticketId -> tickets.id

#33 returns -> #29 orders | orderRawId -> orders.id

#29 orders -> #14 campaigns | campaignId -> campaigns.id

#13 campaignEvents -> #14 campaigns | campaignId -> campaign identity

#47 support_messages -> #41 support_conversations | ticketId -> conversation ID

#09 account_notifications -> #56 users | userId -> users.id

#24 inventoryMovements -> #23 inventory | sku -> inventory.id

Note: Source implementation is documented; deployed configuration and provider connectivity are not verified.

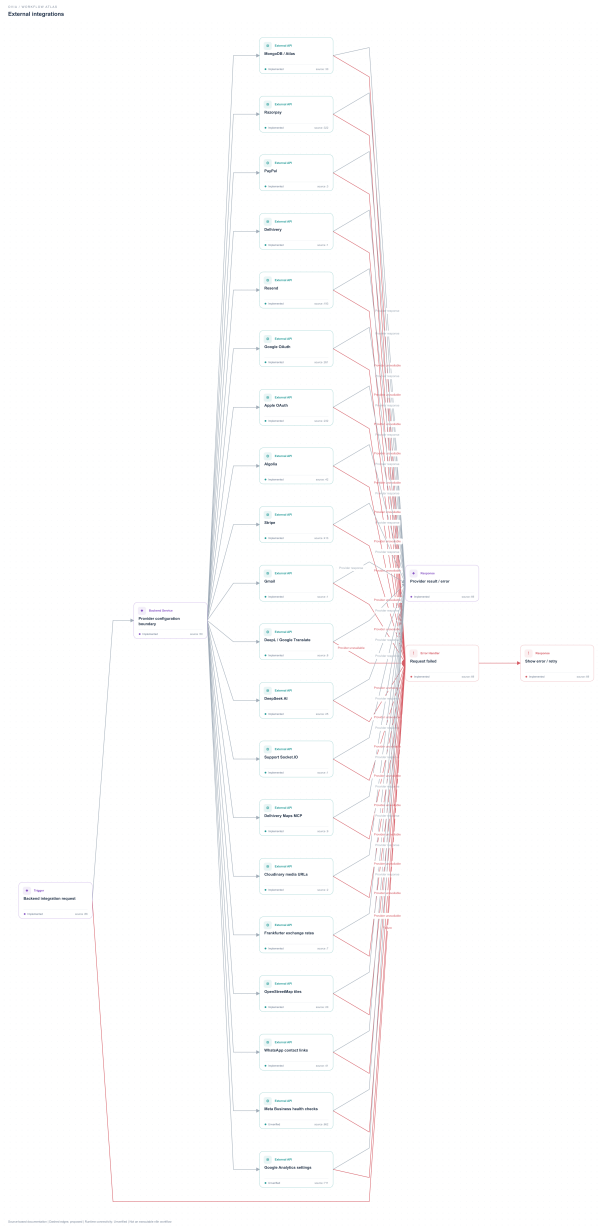
Note: 48 literal collection names were found in queries or index definitions. A collection name in code does not prove that the collection exists in production.

Note: Payments, refunds and shipments are embedded in orders. Wishlist is embedded in users. Categories are matched by catalog fields; no category foreign key is declared.

Note: No Mongoose schemas were found; data models are TypeScript types plus runtime route validation.

External integrations

Architecture overview. Readable node and connection sheets follow.



External integrations

Readable detail sheet 1/7. Node numbers and page references preserve every directed connection.

#01 / Trigger / Implemented

Backend integration request

Service code selects a provider for the requested task.

src/server/integrations-service.ts:88

Next -> #2 (p2); Failure -> #4 (p2)

#02 / Backend Service / Implemented

Provider configuration boundary

Read environment and stored integration settings; no credential values appear in the editor.

src/server/integrations-service.ts:50

Next -> #6 (p3); Next -> #7 (p3); Next -> #8 (p3); Next -> #9 (p4); Next -> #10 (p4); Next -> #11 (p4); Next -> #12 (p4); Next -> #13 (p5); Next -> #14 (p5); Next -> #15 (p5); Next -> #16 (p5); Next -> #17 (p6); Next -> #18 (p6); Next -> #19 (p6); Next...

#03 / Response / Implemented

Provider result / error

Caller receives provider-confirmed data or a route-specific service error.

src/server/http.ts:69

END: no outgoing connection

#04 / Error Handler / Implemented

Request failed

Validation, access, conflict or service errors are returned to the caller. Status and error shape depend on the specific route.

src/server/http.ts:69

Next -> #5 (p3)

External integrations

Readable detail sheet 2/7. Node numbers and page references preserve every directed connection.

#05 / Response / Implemented

Show error / retry

Failure end state. Caller receives the actual error; a retry may be offered by the UI.

src/server/http.ts:69

END: no outgoing connection

#06 / External API / Implemented

MongoDB / Atlas

Native driver and transactional repository. Hosting on Atlas, cluster layout and live connectivity are Unverified.

src/server/db.ts:30

Provider response -> #3 (p2); Provider unavailable -> #4 (p2)

#07 / External API / Implemented

Razorpay

Orders, captured payments, signed webhooks and refunds. Credentials, enabled webhook events and merchant connectivity are Unverified.

src/server/razorpay-checkout.ts:322

Provider response -> #3 (p2); Provider unavailable -> #4 (p2)

#08 / External API / Implemented

PayPal

Sandbox orders, approval, capture and verified webhook. Sandbox endpoint is explicit. Live PayPal settlement and automatic PayPal refunds are Planned.

src/server/paypal.ts:3

Provider response -> #3 (p2); Provider unavailable -> #4 (p2)

External integrations

Readable detail sheet 3/7. Node numbers and page references preserve every directed connection.

#09 / External API / Implemented

Delhivery

Serviceability, shipment, pickup, cancellation and tracking. Courier configuration and live operational readiness are Unverified.

src/lib/shipping/delhivery.ts:1

Provider response -> #3 (p2); Provider unavailable -> #4 (p2)

#10 / External API / Implemented

Resend

Outbound account, order, support and notification email. Sender domain, API configuration and delivered email are Unverified.

src/server/mail.ts:193

Provider response -> #3 (p2); Provider unavailable -> #4 (p2)

#11 / External API / Implemented

Google OAuth

State / PKCE and JWKS-verified identity. Enabled provider configuration is Unverified.

src/server/social-auth.ts:261

Provider response -> #3 (p2); Provider unavailable -> #4 (p2)

#12 / External API / Implemented

Apple OAuth

Identity verification and client-secret signing. Apple keys, enabled provider and runtime readiness are Unverified.

src/server/social-auth.ts:232

Provider response -> #3 (p2); Provider unavailable -> #4 (p2)

External integrations

Readable detail sheet 4/7. Node numbers and page references preserve every directed connection.

#13 / External API / Implemented

Algolia

Search SDK and optional product index synchronization. Index freshness and configured keys are Unverified.

src/lib/algolia.ts:42

Provider response -> #3 (p2); Provider unavailable -> #4 (p2)

#14 / External API / Implemented

Stripe

Legacy test-mode checkout and signed webhook. Checkout is guarded to sk_test_ credentials; live checkout is not claimed.

src/app/api/v1/[[...path]]/route.ts:413

Provider response -> #3 (p2); Provider unavailable -> #4 (p2)

#15 / External API / Implemented

Gmail

OAuth mailbox connection and sync. Mailbox connection, grants and scheduler execution are Unverified.

src/server/support-gmail.ts:1

Provider response -> #3 (p2); Provider unavailable -> #4 (p2)

#16 / External API / Implemented

DeepL / Google Translate

HTTP translation provider selection. Provider selection depends on environment; active configuration is Unverified.

src/app/api/translate/route.ts:8

Provider response -> #3 (p2); Provider unavailable -> #4 (p2)

External integrations

Readable detail sheet 5/7. Node numbers and page references preserve every directed connection.

#17 / External API / Implemented

DeepSeek AI

Owner copilot with constrained provider endpoint. Provider connectivity and model response quality are Unverified.

src/server/ai-copilot.ts:25

Provider response -> #3 (p2); Provider unavailable -> #4 (p2)

#18 / External API / Implemented

Support Socket.IO

Separate realtime support transport. Local support process / deployment transport availability is Unverified.

src/components/use-support-realtime.ts:1

Provider response -> #3 (p2); Provider unavailable -> #4 (p2)

#19 / External API / Implemented

Delhivery Maps MCP

JSON-RPC tools/call for address, search and route tools. Active token and tool availability are Unverified. A hardcoded fallback token exists in source; its value is excluded from documentation.

src/lib/shipping/delhivery-maps.ts:9

Provider response -> #3 (p2); Provider unavailable -> #4 (p2)

#20 / External API / Implemented

Cloudinary media URLs

Validated public-ID image transformation URL helper. The helper creates CDN URLs. Admin uploads currently write to local public/media/uploads; Cloudinary upload API implementation is not claimed.

src/lib/media.ts:2

Provider response -> #3 (p2); Provider unavailable -> #4 (p2)

External integrations

Readable detail sheet 6/7. Node numbers and page references preserve every directed connection.

#21 / External API / Implemented

Frankfurter exchange rates

INR display-rate HTTP fetch with cache. Rates are approximate display conversions; provider failure returns an empty rate list. Runtime availability is Unverified.

src/app/api/v1/rates/route.ts:7

Provider response -> #3 (p2); Provider unavailable -> #4 (p2)

#22 / External API / Implemented

OpenStreetMap tiles

Leaflet browser tile requests for store maps. Public tile rendering is implemented; availability and usage at runtime are Unverified.

src/components/store-map.tsx:20

Provider response -> #3 (p2); Provider unavailable -> #4 (p2)

#23 / External API / Implemented

WhatsApp contact links

Browser deep links to WhatsApp / wa.me. This contact UI opens WhatsApp. It does not prove a server-side automated message-sending API exists.

src/components/whatsapp-widget.tsx:41

Provider response -> #3 (p2); Provider unavailable -> #4 (p2)

#24 / External API / Unverified

Meta Business health checks

Stored settings and Graph API identity checks. Credential-check code exists for WhatsApp/Messenger. Automated message delivery and a verified Messenger webhook are not established by these checks.

src/server/integrations-service.ts:662

Provider response -> #3 (p2); Provider unavailable -> #4 (p2)

External integrations

Readable detail sheet 7/7. Node numbers and page references preserve every directed connection.

#25 / External API / Unverified

Google Analytics settings

Measurement-ID storage and format check. Settings UI exists. No gtag/GA collection call was found in src; automatic GA4 event delivery is Planned, not proven by ID validation.

src/server/integrations-service.ts:711

Provider response -> #3 (p2); Provider unavailable -> #4 (p2)

External integrations

Complete directed connection register / no omitted success, failure or proposed edges

#01 Backend integration request -> #02 Provider configuration boundary | Next

#01 Backend integration request -> #04 Request failed | Failure | FAILURE

#04 Request failed -> #05 Show error / retry | Next | FAILURE

#02 Provider configuration boundary -> #06 MongoDB / Atlas | Next

#06 MongoDB / Atlas -> #03 Provider result / error | Provider response

#06 MongoDB / Atlas -> #04 Request failed | Provider unavailable | FAILURE

#02 Provider configuration boundary -> #07 Razorpay | Next

#07 Razorpay -> #03 Provider result / error | Provider response

#07 Razorpay -> #04 Request failed | Provider unavailable | FAILURE

#02 Provider configuration boundary -> #08 PayPal | Next

#08 PayPal -> #03 Provider result / error | Provider response

#08 PayPal -> #04 Request failed | Provider unavailable | FAILURE

#02 Provider configuration boundary -> #09 Delhivery | Next

#09 Delhivery -> #03 Provider result / error | Provider response

#09 Delhivery -> #04 Request failed | Provider unavailable | FAILURE

#02 Provider configuration boundary -> #10 Resend | Next

#10 Resend -> #03 Provider result / error | Provider response

#10 Resend -> #04 Request failed | Provider unavailable | FAILURE

#02 Provider configuration boundary -> #11 Google OAuth | Next

#11 Google OAuth -> #03 Provider result / error | Provider response

#11 Google OAuth -> #04 Request failed | Provider unavailable | FAILURE

#02 Provider configuration boundary -> #12 Apple OAuth | Next

External integrations

Directed connection register continued

#12 Apple OAuth -> #03 Provider result / error | Provider response

#12 Apple OAuth -> #04 Request failed | Provider unavailable | FAILURE

#02 Provider configuration boundary -> #13 Algolia | Next

#13 Algolia -> #03 Provider result / error | Provider response

#13 Algolia -> #04 Request failed | Provider unavailable | FAILURE

#02 Provider configuration boundary -> #14 Stripe | Next

#14 Stripe -> #03 Provider result / error | Provider response

#14 Stripe -> #04 Request failed | Provider unavailable | FAILURE

#02 Provider configuration boundary -> #15 Gmail | Next

#15 Gmail -> #03 Provider result / error | Provider response

#15 Gmail -> #04 Request failed | Provider unavailable | FAILURE

#02 Provider configuration boundary -> #16 DeepL / Google Translate | Next

#16 DeepL / Google Translate -> #03 Provider result / error | Provider response

#16 DeepL / Google Translate -> #04 Request failed | Provider unavailable | FAILURE

#02 Provider configuration boundary -> #17 DeepSeek AI | Next

#17 DeepSeek AI -> #03 Provider result / error | Provider response

#17 DeepSeek AI -> #04 Request failed | Provider unavailable | FAILURE

#02 Provider configuration boundary -> #18 Support Socket.IO | Next

#18 Support Socket.IO -> #03 Provider result / error | Provider response

#18 Support Socket.IO -> #04 Request failed | Provider unavailable | FAILURE

#02 Provider configuration boundary -> #19 Delhivery Maps MCP | Next

#19 Delhivery Maps MCP -> #03 Provider result / error | Provider response

External integrations

Directed connection register continued

#19 Delhivery Maps MCP -> #04 Request failed | Provider unavailable | FAILURE

#02 Provider configuration boundary -> #20 Cloudinary media URLs | Next

#20 Cloudinary media URLs -> #03 Provider result / error | Provider response

#20 Cloudinary media URLs -> #04 Request failed | Provider unavailable | FAILURE

#02 Provider configuration boundary -> #21 Frankfurter exchange rates | Next

#21 Frankfurter exchange rates -> #03 Provider result / error | Provider response

#21 Frankfurter exchange rates -> #04 Request failed | Provider unavailable | FAILURE

#02 Provider configuration boundary -> #22 OpenStreetMap tiles | Next

#22 OpenStreetMap tiles -> #03 Provider result / error | Provider response

#22 OpenStreetMap tiles -> #04 Request failed | Provider unavailable | FAILURE

#02 Provider configuration boundary -> #23 WhatsApp contact links | Next

#23 WhatsApp contact links -> #03 Provider result / error | Provider response

#23 WhatsApp contact links -> #04 Request failed | Provider unavailable | FAILURE

#02 Provider configuration boundary -> #24 Meta Business health checks | Next

#24 Meta Business health checks -> #03 Provider result / error | Provider response

#24 Meta Business health checks -> #04 Request failed | Provider unavailable | FAILURE

#02 Provider configuration boundary -> #25 Google Analytics settings | Next

#25 Google Analytics settings -> #03 Provider result / error | Provider response

#25 Google Analytics settings -> #04 Request failed | Provider unavailable | FAILURE

Note: Source implementation is documented; deployed configuration and provider connectivity are not verified.

Note: All listed adapters have code evidence. Runtime configuration is Unverified because this task does not read production secrets or call live providers. No integration is labeled Configured without configuration evidence.

MongoDB connection & Atlas boundary

Architecture overview. Readable node and connection sheets follow.



MongoDB connection & Atlas boundary

Readable detail sheet 1/2. Node numbers and page references preserve every directed connection.

#01 / Trigger / Implemented

Database operation requested

Server code requests a shared MongoDB connection.

src/server/db.ts:30

Next -> #2 (p2)

#02 / IF / Switch / Implemented

MONGODB_URI supplied?

Absence rejects live operations; no credentials are displayed.

src/server/db.ts:31

Yes / accepted -> #3 (p2); No / rejected -> #7 (p3)

#03 / Database Query / Implemented

Shared native-driver pool

MongoClient.connect, maxPoolSize 10 and serverSelectionTimeoutMS 6000 are actual code settings.

src/server/db.ts:34

Next -> #4 (p2)

#04 / External API / Unverified

Hosted cluster / Atlas boundary

MongoDB hosting, Atlas cluster tier, region, replica placement and network access rules cannot be inferred from the connection code.

src/server/db.ts:33

Next -> #5 (p3)

MongoDB connection & Atlas boundary

Readable detail sheet 2/2. Node numbers and page references preserve every directed connection.

#05 / Database Query / Implemented

Database selection and transaction

MONGODB_DB override or ovia default; writes can share ClientSession.withTransaction.

src/server/db.ts:103

Next -> #6 (p3)

#06 / Response / Implemented

Repository result returned

Stored records or a surfaced database error return to business services.

src/server/db.ts:44

END: no outgoing connection

#07 / Error Handler / Implemented

Request failed

Validation, access, conflict or service errors are returned to the caller. Status and error shape depend on the specific route.

src/server/http.ts:69

Next -> #8 (p3)

#08 / Response / Implemented

Show error / retry

Failure end state. Caller receives the actual error; a retry may be offered by the UI.

src/server/http.ts:69

END: no outgoing connection

MongoDB connection & Atlas boundary

Complete directed connection register / no omitted success, failure or proposed edges

#01 Database operation requested -> #02 MONGODB_URI supplied? | Next

#02 MONGODB_URI supplied? -> #03 Shared native-driver pool | Yes / accepted

#03 Shared native-driver pool -> #04 Hosted cluster / Atlas boundary | Next

#04 Hosted cluster / Atlas boundary -> #05 Database selection and transaction | Next

#05 Database selection and transaction -> #06 Repository result returned | Next

#02 MONGODB_URI supplied? -> #07 Request failed | No / rejected | FAILURE

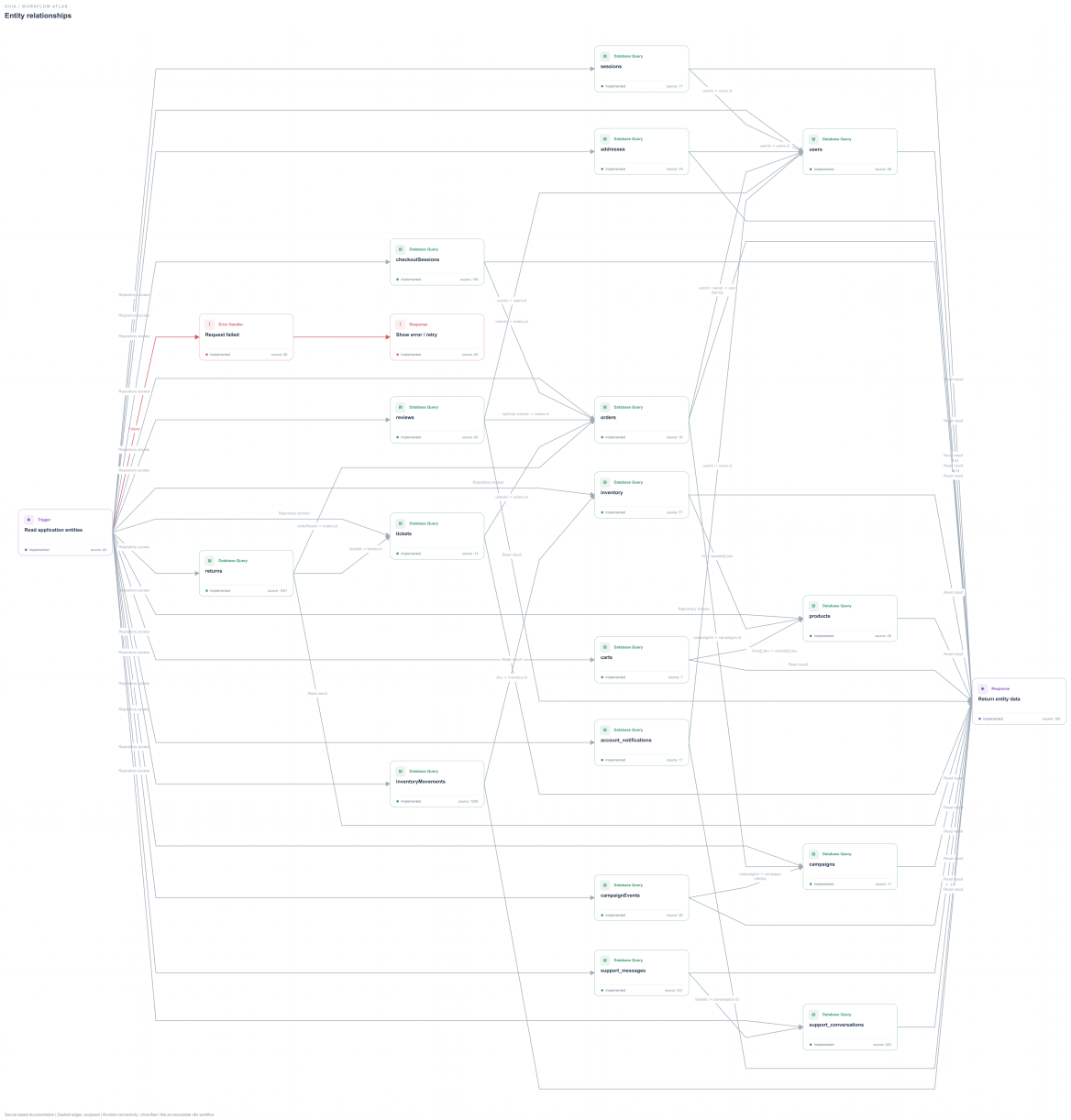
#07 Request failed -> #08 Show error / retry | Next | FAILURE

Note: Source implementation is documented; deployed configuration and provider connectivity are not verified.

Note: The hosted-cluster node is a trust boundary, not proof of an Atlas deployment. No private network, replica topology or production configuration has been invented.

Entity relationships

Architecture overview. Readable node and connection sheets follow.



Entity relationships

Readable detail sheet 1/6. Node numbers and page references preserve every directed connection.

#01 / Trigger / Implemented

Read application entities

Entity and reference view derived from actual collection access and reference fields.

src/server/db.ts:44

Failure -> #20 (p6); Repository access -> #2 (p2); Repository access -> #3 (p2); Repository access -> #4 (p2); Repository access -> #5 (p3); Repository access -> #6 (p3); Repository access -> #7 (p3); Repository access -> #8 (p3); Repository...

#02 / Database Query / Implemented

sessions

Application entity stored in sessions. Relationship edges identify actual reference fields and imply no MongoDB foreign-key constraint.

src/app/api/admin/security/route.ts:17

Read result -> #19 (p6); userId -> users.id -> #3 (p2)

#03 / Database Query / Implemented

users

Application entity stored in users. Relationship edges identify actual reference fields and imply no MongoDB foreign-key constraint.

src/app/api/admin/customers/route.ts:59

Read result -> #19 (p6)

#04 / Database Query / Implemented

addresses

Application entity stored in addresses. Relationship edges identify actual reference fields and imply no MongoDB foreign-key constraint.

src/app/api/v1/account/addresses/[[...id]]/route.ts:19

Read result -> #19 (p6); userId -> users.id -> #3 (p2)

Entity relationships

Readable detail sheet 2/6. Node numbers and page references preserve every directed connection.

#05 / Database Query / Implemented

orders

Application entity stored in orders. Relationship edges identify actual reference fields and imply no MongoDB foreign-key constraint.

src/app/api/admin/orders/[id]/document/route.ts:16

Read result -> #19 (p6); userId / owner -> user identity -> #3 (p2); campaignId -> campaigns.id -> #13 (p5)

#06 / Database Query / Implemented

checkoutSessions

Application entity stored in checkoutSessions. Relationship edges identify actual reference fields and imply no MongoDB foreign-key constraint.

src/server/commerce.ts:136

Read result -> #19 (p6); orderId -> orders.id -> #5 (p3)

#07 / Database Query / Implemented

inventory

Application entity stored in inventory. Relationship edges identify actual reference fields and imply no MongoDB foreign-key constraint.

src/app/api/v1/orders/[id]/return/route.ts:77

Read result -> #19 (p6); id -> variants[].sku -> #8 (p3)

#08 / Database Query / Implemented

products

Application entity stored in products. Relationship edges identify actual reference fields and imply no MongoDB foreign-key constraint.

src/app/api/admin/product-taxonomy/route.ts:30

Read result -> #19 (p6)

Entity relationships

Readable detail sheet 3/6. Node numbers and page references preserve every directed connection.

#09 / Database Query / Implemented

carts

Application entity stored in carts. Relationship edges identify actual reference fields and imply no MongoDB foreign-key constraint.

src/server/account-sign-in.ts:7

Read result -> #19 (p6); lines[].sku -> variants[].sku -> #8 (p3)

#10 / Database Query / Implemented

reviews

Application entity stored in reviews. Relationship edges identify actual reference fields and imply no MongoDB foreign-key constraint.

src/app/api/v1/reviews/route.ts:63

Read result -> #19 (p6); userId -> users.id -> #3 (p2); optional orderId -> orders.id -> #5 (p3)

#11 / Database Query / Implemented

tickets

Application entity stored in tickets. Relationship edges identify actual reference fields and imply no MongoDB foreign-key constraint.

src/app/api/v1/account/activity/route.ts:14

Read result -> #19 (p6); orderId -> orders.id -> #5 (p3)

#12 / Database Query / Implemented

returns

Application entity stored in returns. Relationship edges identify actual reference fields and imply no MongoDB foreign-key constraint.

src/server/admin-service.ts:1401

Read result -> #19 (p6); ticketId -> tickets.id -> #11 (p4); orderRawId -> orders.id -> #5 (p3)

Entity relationships

Readable detail sheet 4/6. Node numbers and page references preserve every directed connection.

#13 / Database Query / Implemented

campaigns

Application entity stored in campaigns. Relationship edges identify actual reference fields and imply no MongoDB foreign-key constraint.

src/server/campaigns.ts:11

Read result -> #19 (p6)

#14 / Database Query / Implemented

campaignEvents

Application entity stored in campaignEvents. Relationship edges identify actual reference fields and imply no MongoDB foreign-key constraint.

src/server/campaigns.ts:20

Read result -> #19 (p6); campaignId -> campaign identity -> #13 (p5)

#15 / Database Query / Implemented

support_messages

Application entity stored in support_messages. Relationship edges identify actual reference fields and imply no MongoDB foreign-key constraint.

src/app/api/v1/care/[[...path]]/route.ts:231

Read result -> #19 (p6); ticketId -> conversation ID -> #16 (p5)

#16 / Database Query / Implemented

support_conversations

Application entity stored in support_conversations. Relationship edges identify actual reference fields and imply no MongoDB foreign-key constraint.

src/app/api/v1/care/[[...path]]/route.ts:453

Read result -> #19 (p6)

Entity relationships

Readable detail sheet 5/6. Node numbers and page references preserve every directed connection.

#17 / Database Query / Implemented

account_notifications

Application entity stored in account_notifications. Relationship edges identify actual reference fields and imply no MongoDB foreign-key constraint.

src/app/api/v1/account/activity/route.ts:17

Read result -> #19 (p6); userId -> users.id -> #3 (p2)

#18 / Database Query / Implemented

inventoryMovements

Application entity stored in inventoryMovements. Relationship edges identify actual reference fields and imply no MongoDB foreign-key constraint.

src/server/admin-service.ts:1206

Read result -> #19 (p6); sku -> inventory.id -> #7 (p3)

#19 / Response / Implemented

Return entity data

Read results return to the application.

src/server/db.ts:100

END: no outgoing connection

#20 / Error Handler / Implemented

Request failed

Validation, access, conflict or service errors are returned to the caller. Status and error shape depend on the specific route.

src/server/http.ts:69

Next -> #21 (p7)

Entity relationships

Readable detail sheet 6/6. Node numbers and page references preserve every directed connection.

#21 / Response / Implemented

Show error / retry

Failure end state. Caller receives the actual error; a retry may be offered by the UI.

src/server/http.ts:69

END: no outgoing connection

Entity relationships

Complete directed connection register / no omitted success, failure or proposed edges

#01 Read application entities -> #20 Request failed | Failure | FAILURE

#20 Request failed -> #21 Show error / retry | Next | FAILURE

#01 Read application entities -> #02 sessions | Repository access

#02 sessions -> #19 Return entity data | Read result

#01 Read application entities -> #03 users | Repository access

#03 users -> #19 Return entity data | Read result

#01 Read application entities -> #04 addresses | Repository access

#04 addresses -> #19 Return entity data | Read result

#01 Read application entities -> #05 orders | Repository access

#05 orders -> #19 Return entity data | Read result

#01 Read application entities -> #06 checkoutSessions | Repository access

#06 checkoutSessions -> #19 Return entity data | Read result

#01 Read application entities -> #07 inventory | Repository access

#07 inventory -> #19 Return entity data | Read result

#01 Read application entities -> #08 products | Repository access

#08 products -> #19 Return entity data | Read result

#01 Read application entities -> #09 carts | Repository access

#09 carts -> #19 Return entity data | Read result

#01 Read application entities -> #10 reviews | Repository access

#10 reviews -> #19 Return entity data | Read result

#01 Read application entities -> #11 tickets | Repository access

#11 tickets -> #19 Return entity data | Read result

Entity relationships

Directed connection register continued

#01 Read application entities -> #12 returns | Repository access

#12 returns -> #19 Return entity data | Read result

#01 Read application entities -> #13 campaigns | Repository access

#13 campaigns -> #19 Return entity data | Read result

#01 Read application entities -> #14 campaignEvents | Repository access

#14 campaignEvents -> #19 Return entity data | Read result

#01 Read application entities -> #15 support_messages | Repository access

#15 support_messages -> #19 Return entity data | Read result

#01 Read application entities -> #16 support_conversations | Repository access

#16 support_conversations -> #19 Return entity data | Read result

#01 Read application entities -> #17 account_notifications | Repository access

#17 account_notifications -> #19 Return entity data | Read result

#01 Read application entities -> #18 inventoryMovements | Repository access

#18 inventoryMovements -> #19 Return entity data | Read result

#02 sessions -> #03 users | userId -> users.id

#04 addresses -> #03 users | userId -> users.id

#05 orders -> #03 users | userId / owner -> user identity

#06 checkoutSessions -> #05 orders | orderId -> orders.id

#07 inventory -> #08 products | id -> variants[].sku

#09 carts -> #08 products | lines[].sku -> variants[].sku

#10 reviews -> #03 users | userId -> users.id

#10 reviews -> #05 orders | optional orderId -> orders.id

Entity relationships

Directed connection register continued

#11 tickets -> #05 orders | orderId -> orders.id

#12 returns -> #11 tickets | ticketId -> tickets.id

#12 returns -> #05 orders | orderRawId -> orders.id

#05 orders -> #13 campaigns | campaignId -> campaigns.id

#14 campaignEvents -> #13 campaigns | campaignId -> campaign identity

#15 support_messages -> #16 support_conversations | ticketId -> conversation ID

#17 account_notifications -> #03 users | userId -> users.id

#18 inventoryMovements -> #07 inventory | sku -> inventory.id

Note: Source implementation is documented; deployed configuration and provider connectivity are not verified.

Note: Reference edges represent application-level lookup relationships. Repository access/result edges preserve diagram entry and exit, not additional entity relationships.

Note: orders embeds paymentInfo, refunds[], shipping and logisticsOperations; users embeds accountWishlist. No nonexistent separate entity is introduced.